

Project Initialization and Planning Phase

Date	15 March 2026
Team ID	xxxxxx
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to classify a region's Human Development Index (HDI) category using machine learning, boosting efficiency and accuracy in development assessments. It tackles the inefficiency of manual indicator analysis, promising faster insights, consistent results, and better-informed policy decisions. Key features include a machine learning-based HDI classification model and real-time prediction via a web application.

Project Overview	
Objective	The primary objective is to build a machine learning model that predicts a region's HDI category (Low, Medium, High, or Very High Human Development) from socio-economic indicators, replacing slow manual computation with an instant, consistent classification.
Scope	The project covers data collection from UNDP/World Bank sources, preprocessing of HDI indicator data, training and evaluating a classification model, and deploying it as a Flask web application for real-time predictions.
Problem Statement	
Description	Computing a region's HDI category manually from several weighted indicators (life expectancy, education index, GNI per capita) is slow, error-prone, and inaccessible to non-technical users, limiting timely development analysis.
Impact	Solving this will result in faster, more consistent HDI assessments available to policymakers, NGOs and researchers, enabling more timely and data-driven decisions on resource allocation and development priorities.
Proposed Solution	
Approach	Train a supervised machine learning classification model on historical HDI indicator data labeled with known development categories, then serve predictions through a simple Flask web interface.
Key Features	- Implementation of a machine learning-based HDI category classification model. - Real-time prediction via a web form: enter indicators, get an instant category result (e.g. HIGH HUMAN DEVELOPMENT). - Clear, human-readable output with error handling for invalid inputs.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU (T4 GPU on Google Colab for training)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	5-10 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, joblib, matplotlib
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm
Data		
Data	Source, size, format	UNDP Human Development Report dataset (CSV); World Bank open data indicators (CSV)